Grocery Delivery System

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Problem Statement

While buying groceries may seem simple, for many people it has become a frustrating task. People are busy, stores are crowded, and many do not have the time to shop in person. Older customers may also struggle due to mobility limitations or long wait times.

This project addresses these challenges by providing a simple and efficient grocery delivery system that allows users to browse products, place orders, and track deliveries online with minimal effort.

System Requirements

The system will allow users to:

- Register and log into an account
- Browse grocery products
- Search for products by name
- Filter products by category
- Add items to a shopping cart
- Update item quantities in the cart
- Remove items from the cart
- Place an order
- Enter delivery information (name and address)
- Select a delivery time slot
- View order history
- Track order status automatically (Preparing → Out for Delivery → Delivered)

Functional Requirements

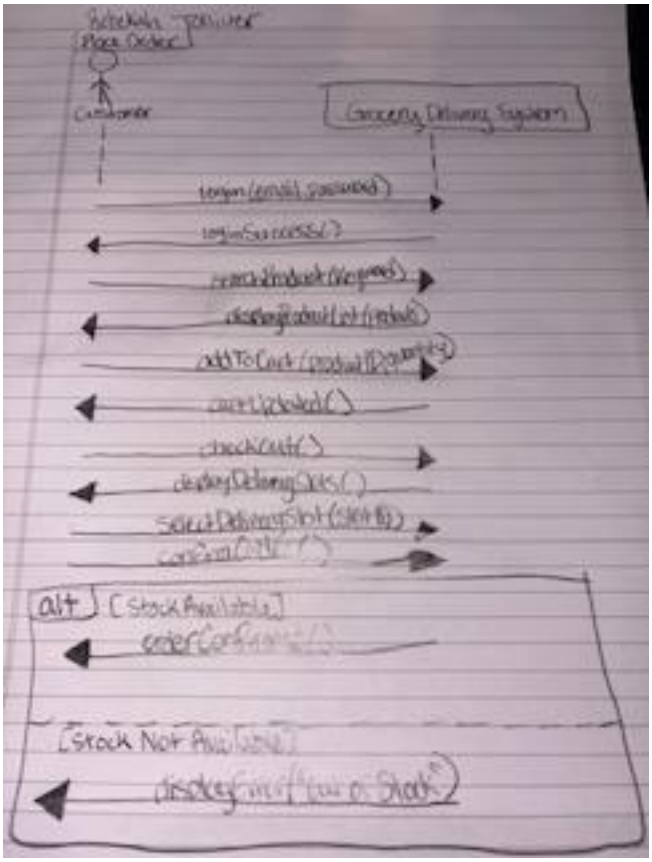
No.	Priority	Description
REQ-1	High	The system shall allow users to register an account
REQ-2	High	The system shall allow users to log in
REQ-3	High	Users shall be able to browse products
REQ-4	High	Users shall be able to search products by name
REQ-5	High	Users shall be able to filter products by category

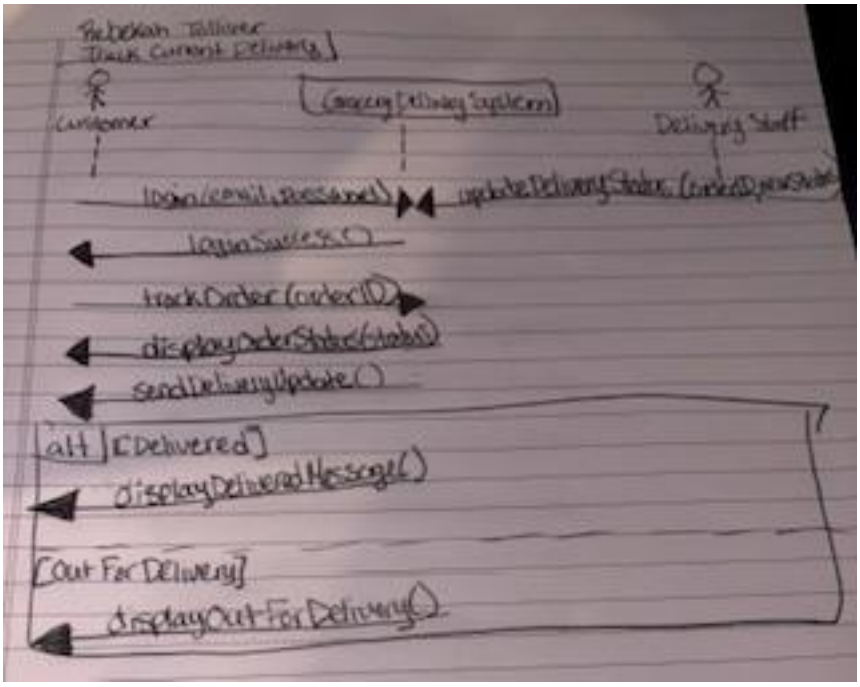
REQ-6	High	Users shall be able to add items to a cart
REQ-7	High	Users shall be able to update item quantities
REQ-8	High	Users shall be able to remove items from the cart
REQ-9	High	Users shall be able to place an order
REQ-10	High	Users shall be able to enter delivery details
REQ-11	Medium	Users shall be able to select a delivery time slot
REQ-12	High	Users shall be able to view order history
REQ-13	High	The system shall automatically update delivery status

Nonfunctional Requirements

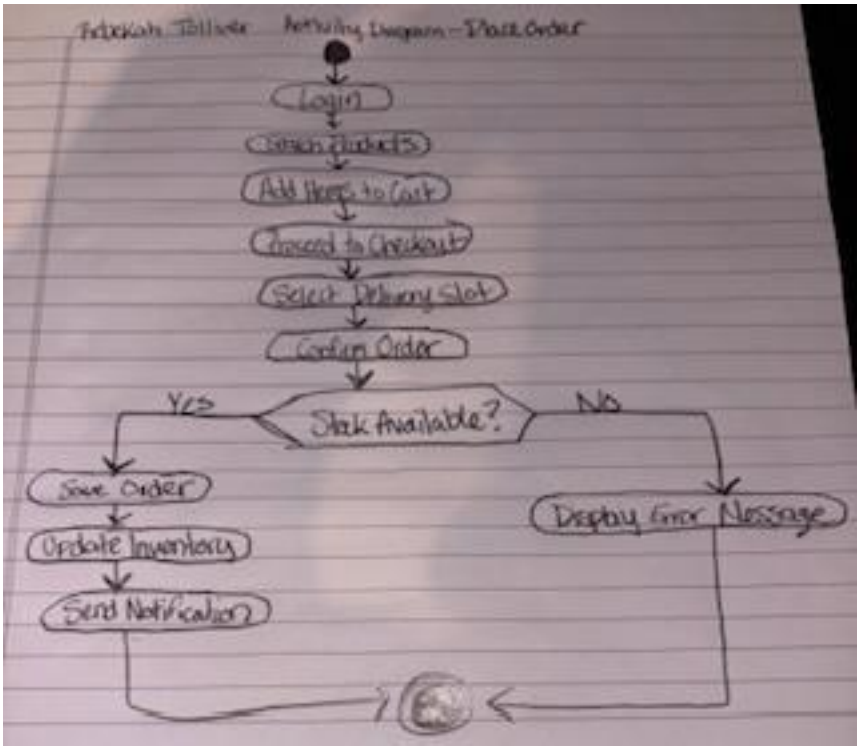
No.	Category	Priority	Description
NFR-1	Usability	High	The system shall be simple and easy to use
NFR-2	Reliability	High	The system shall correctly store cart and order data
NFR-3	Performance	Medium	The system shall respond quickly to user input
NFR-4	Functionality	High	The system shall update delivery status automatically
NFR-5	Supportability	Medium	The system shall allow future improvements

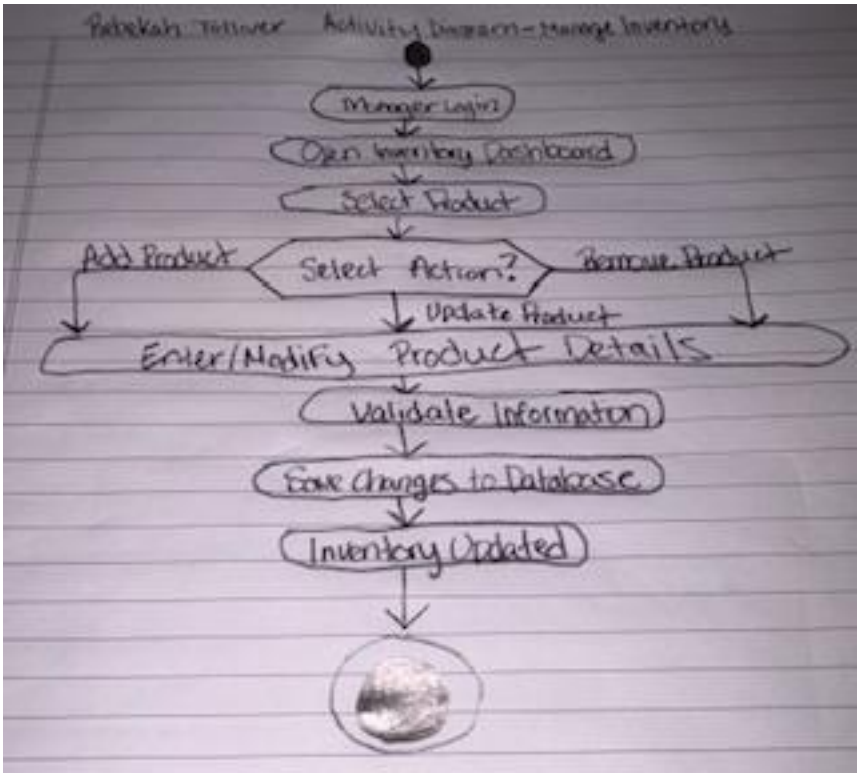
System Sequence Diagrams





Activity Diagrams





User Interface Specifications

Selected Use Cases:

- Register Account
- Place Order
- Track Delivery
- View Order History

1. Register Account

Input:

- Name, email, password

Output:

- Success or error message

Navigation:
Login Page, Register, Confirmation

2. Place Order

Steps:

- Login
- Search or filter products
- Add items to cart
- Adjust quantities
- Click checkout
- Enter name and address
- Select delivery time
- Confirm order

Output:

- “Order placed successfully”

3. Track Delivery

Steps:

- View order history
- Observe status updates

Output:

- Status changes automatically:
- Preparing
- Out for Delivery
- Delivered

4. View Order History

Steps:

- Scroll to order history section

Output:

- List of past orders with total and status

Project Plan

Technologies Used:

- HTML
- CSS
- JavaScript
- Browser Local Storage

Hardware:

- PC or laptop

Development Timeline:

- Weeks 1–2: Requirements and setup
- Weeks 3–4: Authentication system
- Weeks 5–7: Product browsing and cart
- Week 8: Midterm demo
- Weeks 9–11: Checkout and delivery system
- Weeks 12–14: UI improvements and testing
- Week 15: Final demo

References

- Course materials

- HTML, CSS, JavaScript documentation